

# Sukkeun KIM, PhD

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## Bio

**Sukkeun Kim** is an aerospace engineer specialising in nonlinear Bayesian filtering and navigation systems. He bridges the gap between advanced mathematical theory and real-world implementation by developing and testing multiple estimation algorithms including particle flow and Gaussian mixture filters. He earned his BSc and MSc from Pusan National University and his PhD from Cranfield University. His research experience includes roles as a visiting student researcher at NASA's Jet Propulsion Laboratory (JPL) and the KAIST Graduate School of Mobility, focusing on applications in multi-target tracking, autonomous robotic localisation and space navigation systems.

## SKILLS AND INTERESTS

**Technical Expertise:** Navigation/Localisation, State Estimation, Nonlinear Filtering (EKF, UKF, PF, GMF, PFF), Multi-Target Tracking (JPDA), Sensor Fusion, Computer Vision, SLAM

**Languages & Tools:** C++, Python, MATLAB/Simulink, Julia, Linux Ubuntu, ROS 1/2, OpenCV, Jetson TX2/Nano

## TECHNICAL AND RESEARCH EXPERIENCE

### Intelligent Sensor-Actuator-Systems (ISAS), KIT

Postdoctoral Researcher (Wissenschaftlicher Mitarbeiter)

Karlsruhe, Germany

September 2025–September 2026

- **Nonlinear Filtering:** Developed advanced particle flow-based filters (PFFs) for applications to nonlinear systems
- **Localisation Curriculum Design:** Appointed to design and lecture a full-semester master's course on *Localization of Mobile Agents (LMA)*, managing technical transfer of Bayesian estimation, sensor fusion and localisation frameworks

### CCS Graduate School of Mobility, KAIST

Visiting Researcher

Daejeon, Republic of Korea

February 2025–August 2025

- PhD thesis project and supervision of graduate school students

### NASA Jet Propulsion Laboratory (JPL)

JPL Visiting Student Research Program (JVS RP)

Pasadena, USA

September 2024–February 2025

- **Autonomous Space Navigation Pipelines:** Developed and evaluated advanced Bayesian estimation for autonomous robotic signal source localisation in unknown and unstructured environment
- **Algorithm Frameworks:** Implemented radio signal source localisation architecture on Linux Ubuntu and ROS 1

### Cranfield University

PhD Researcher & Part-time Research Assistant

Cranfield, UK

September 2021–January 2026

- **Nonlinear Filtering Architecture:** Designed, simulated and deployed hybrid Bayesian estimation architectures combined with AI approaches for target tracking and source localisation in highly nonlinear and complex environments
- **Algorithm Development:** Programmed modular state-estimation pipelines for nonlinear Bayesian estimation and vision-based multi-class, multi-object tracking frameworks based on CNN and JPDA
- **Technical Mentor:** Guided laboratory implementation sessions using MATLAB, instructing students on practical sensor fusion module integration and Master's thesis supervision
- **RA Project (Part-time, October 2023–August 2024):** Conducted a study on operational procedures for UAM/AAM take-off and landing at vertiports adjacent to airports

### NEXTFOAM

Research Intern

Seoul, Republic of Korea

March 2021–July 2021

- **Vision Algorithm:** Developed and evaluated a vision-based tracking algorithm and stereo vision system
- **Embedded Systems:** Implemented the architecture C++ on embedded system (Jetson Nano with camera)

### Pusan National University

Part-time Research Assistant

Busan, Republic of Korea

August 2019–December 2020

- **Embedded Flight Software:** Engineered vision-based relative position measurement and tracking software in C++ deployed directly onto embedded NVIDIA Jetson TX2 hardware
- **Verification & Validation (V&V):** Successfully completed hardware-in-the-loop validation, collaborative UAV-UGV team field operations and live flight test verification
- **Relative Navigation Systems:** Designed deep-learning-based relative navigation software validated via light sport aircraft flight test data and collaboration with research institutes

EDUCATION

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| <b>Cranfield University</b><br><i>PhD in Aerospace, Faculty of Engineering and Applied Sciences</i> <ul style="list-style-type: none"><li>Thesis (Supervisors: Professor Hyo-Sang Shin and Professor Ivan Petrunin):<br/>Hybrid Bayesian Estimation Approaches for Tracking and Localisation in Complex Environments</li></ul> <i>Visiting Student, Centre for Autonomous and Cyber-Physical Systems</i> <ul style="list-style-type: none"><li>Autonomous Vehicles Control System</li><li>Guidance and Navigation for Autonomous Systems</li></ul> | Cranfield, UK<br>January 2026<br><br><br><br><br>December 2015–February 2016 |
| <b>Pusan National University</b><br><i>MSc in Aerospace Engineering (CGPA: 4.19/4.5)</i> <ul style="list-style-type: none"><li>Thesis (Supervisor: Professor Daewoo Lee):<br/>Vision-based Relative Navigation with CNN-aided 3D Model Matching for Non-cooperative Airplanes</li></ul> <i>BSc in Aerospace Engineering (CGPA: 3.77/4.5)</i> <ul style="list-style-type: none"><li>Thesis (Supervisor: Professor Daewoo Lee):<br/>Relative Position/Attitude Measurement of Leader Aircraft in Formation Flight using MBT</li></ul>                | Busan, Republic of Korea<br>February 2021<br><br><br><br>February 2019       |

SELECTED PEER-REVIEWED PUBLICATIONS (FULL LIST: )

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| <b>Out of 6 Journals, 2 Preprints, 5 Conferences including IEEE, Information Fusion and IFAC</b> <ul style="list-style-type: none"><li>P2: <b>S. Kim</b>, D. Frisch, and U. D. Hanebeck, “The ultimate hitchhiker’s guide to particle flow-based filters: A survey,” in <i>preparation</i>, 2026</li><li>J6: <b>S. Kim</b>, M. Sun, I. Petrunin, and H.-S. Shin, “DBSCAN-based particle Gaussian mixture filters,” <i>Digital Signal Processing</i>, vol. 168, p. 105546, 2026</li><li>J4: <b>S. Kim</b>, I. Petrunin, and H.-S. Shin, “AFJPDA: A multiclass multi-object tracking with appearance feature-aided joint probabilistic data association,” <i>Journal of Aerospace Information Systems</i>, vol. 21, no. 4, pp. 294–304, 2024</li><li>J3: <b>S. Kim</b>, J. Kim, J. Park, and D. Lee, “Vision-based pose estimation of fixed-wing aircraft using you only look once and perspective-n-points,” <i>Journal of Aerospace Information Systems</i>, vol. 18, no. 9, pp. 659–664, 2021</li><li>C3: <b>S. Kim</b>, S. Moon, I. Petrunin, H.-S. Shin, and S. Khattak, “Autonomous robotic radio source localization via a novel gaussian mixture filtering approach,” in <i>2025 28th International Conference on Information Fusion (FUSION)</i>, (Rio de Janeiro, Brazil), pp. 1–8, July 2025</li></ul> |  |
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LEADERSHIP AND TECHNICAL COMMUNICATION

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| <b>Lecturer &amp; Teaching Assistant (KIT &amp; Cranfield University)</b> <ul style="list-style-type: none"><li>Appointed to design and deliver a full-semester Master’s course on <b>Localization of Mobile Agents (LMA)</b> at KIT. Invited to deliver special guest lectures on <b>State Estimation and Multi-target Tracking</b> for Cranfield’s AVDC modules</li></ul> |  |
| <b>Technical Mentorship &amp; Hardware Facilitation</b> <ul style="list-style-type: none"><li>Supervised 2 Bachelor student theses focused on progressive filtering, localisation and multi-sensor fusion. Mentored postgraduate and undergraduate students during practical MATLAB sensor fusion implementations and embedded system hardware deployments</li></ul>        |  |

HONOURS AND AWARDS

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| <b>Honours</b> <ul style="list-style-type: none"><li>Fully-funded PhD Studentship, Cranfield University</li><li>Scholarship for Overseas Students, PNU Alumni Professors’ Association</li><li>Scholarship for Academic Excellence (full or partial)</li></ul> | September 2021–March 2025<br>July 2021<br>September 2015–February 2019                                                                                                                                  |
| <b>Awards</b> <ul style="list-style-type: none"><li>2022 ICNS Best Student Paper Award</li><li>KSAA Student Paper Competition</li><li>KSAS Student Paper Competition</li><li>Korea Robot Aircraft Competition</li><li>Drone Challenge Korea</li></ul>         | 2 <sup>nd</sup> Runner up in 2022<br>1 <sup>st</sup> Prize in 2020<br>1 <sup>st</sup> Prize in 2019<br>1 <sup>st</sup> Prize in 2018 and 3 <sup>rd</sup> Prize in 2019<br>3 <sup>rd</sup> Prize in 2018 |

SKILLS AND INTERESTS

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| <b>Languages:</b> Korean (Native/Fluent), English (Full Professional Proficiency), German (Basic)                 |  |
| <b>Applications:</b> SolidWorks, Auto CAD, Microsoft Word/Excel/Visio and Adobe Photoshop                         |  |
| <b>Personal interests:</b> Space, photography and running (Marathon finisher: MK 2024 Seoul 2025 and Berlin 2026) |  |